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DIGITAL LIBRARY MANAGEMENT SYSTEM

Software Requirements Specification (SRS)

&

High Level Design (HLD) Document

Prepared For	Nagpur Municipal Corporation, IT Department
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PART I

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose of the Document

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements for the Digital Library Management System (DLMS) proposed by Cluematrix Technologies Private Limited for Nagpur Municipal Corporation (NMC). This document is intended to serve as the basis for mutual understanding between the client (NMC) and the development team (Cluematrix) regarding the scope, behaviour, and constraints of the proposed system. All stakeholders shall treat this document as the authoritative reference throughout the project lifecycle.

1.2 Scope of the System

The Digital Library Management System (DLMS) is a comprehensive web-based application which shall manage and digitise all library-related operations across libraries under the purview of Nagpur Municipal Corporation. The system shall cater to the following broad functional areas:

- Administration and governance of all NMC library branches from a centralised dashboard
- Digital cataloguing and ISBN-based management of books and study materials
- Citizen registration, membership management, and borrowing privileges
- Book issue, return, reservation, and fine computation
- Digital content repository comprising eBooks, research papers, and video resources
- Online self-service portal for citizens to search, reserve, and access library resources
- Automated notification system via SMS and Email
- Comprehensive MIS reporting and analytics for library administrators

1.3 Definitions, Abbreviations, and Acronyms

Term / Abbreviation	Full Form / Meaning
DLMS	Digital Library Management System
NMC	Nagpur Municipal Corporation
SRS	Software Requirements Specification
HLD	High Level Design

Term / Abbreviation	Full Form / Meaning
ISBN	International Standard Book Number
API	Application Programming Interface
RBAC	Role Based Access Control
UI	User Interface
MIS	Management Information System
OTP	One Time Password
SSL	Secure Sockets Layer
JWT	JSON Web Token
PDF	Portable Document Format
SMS	Short Message Service
UAT	User Acceptance Testing
VAPT	Vulnerability Assessment and Penetration Testing

1.4 Intended Audience

- NMC IT Department – for review and approval of requirements
- Cluematrix Development and QA Team – for design, development, and testing
- NMC Library Administrators and Librarians – for UAT participation and feedback
- Project Managers – for project planning and milestone tracking

2. Overall System Description

2.1 Product Perspective

The DLMS shall function as a standalone web-based multi-tier application accessible via standard web browsers on desktops, laptops, tablets, and mobile devices. It shall be hosted on a cloud or on-premise server as per the infrastructure policy of Nagpur Municipal Corporation. The system shall interact with external SMS and Email gateway providers for sending automated notifications to citizens and library staff.

The system shall replace existing manual and paper-based library processes across all NMC libraries with a unified digital platform, thereby eliminating redundancy, reducing human error, and significantly improving the overall efficiency and transparency of library services extended to the citizens of Nagpur.

2.2 User Classes and Characteristics

User Role	Description	Access Level
Super Administrator	NMC IT Department officials responsible for overall system governance, configuration, and monitoring	Full system access across all branches
Library Administrator	In-charge officers managing individual library branches under NMC	Branch-level administration
Librarian	Day-to-day library staff handling book issuance, return, cataloguing, and member queries	Operational access within assigned branch
Citizen / Member	Registered citizens of Nagpur who have enrolled as library members	Self-service portal, digital content access
Guest User	Non-registered visitors who wish to browse the public book catalogue	Read-only catalogue search

2.3 Operating Environment

- Server: Linux-based cloud or on-premise server (Ubuntu 22.04 LTS recommended) with Java Spring Boot backend
- Database: MySQL 8.x or PostgreSQL 15 (relational DBMS)
- Client: Any modern web browser – Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari
- Network: Internet connectivity for cloud deployment; Local Area Network (LAN) for on-premise setup
- Mobile: Fully responsive web interface accessible on Android and iOS devices without need for a separate mobile application

2.4 Assumptions and Dependencies

- NMC shall provide adequate server infrastructure or procure cloud hosting as per their internal IT policy prior to the deployment phase
- Citizen mobile numbers and Email addresses shall be collected mandatorily at the time of membership registration for notification purposes
- Existing library branch data, book inventories, and member records (if any) shall be provided by NMC in structured format during the data migration phase
- A valid SSL certificate shall be procured and installed on the production server prior to go-live
- SMS and Email gateway credentials and associated costs shall be arranged by NMC or by Cluematrix as mutually agreed upon during project initiation

3. Functional Requirements

3.1 Admin Dashboard Module

3.1.1 Description

The Admin Dashboard shall serve as the nerve centre of the entire DLMS. Super Administrators and Library Administrators shall use this module for governance, monitoring, and configuration of the system across all branches.

3.1.2 Functional Requirements

Req. ID	Requirement Description	Priority
FR-AD-01	The system shall display a real-time summary dashboard showing total members, total books, books issued today, pending returns, and overdue items across all branches	High
FR-AD-02	The system shall allow Super Admin to add, edit, activate, or deactivate library branches	High
FR-AD-03	The system shall allow Super Admin to create and manage user accounts for Library Administrators and Librarians with appropriate role assignments	High
FR-AD-04	The system shall implement Role Based Access Control (RBAC) ensuring each user can access only those modules relevant to their designated role	High
FR-AD-05	The system shall provide branch-wise performance comparison reports in graphical format (bar charts, pie charts)	Medium
FR-AD-06	The system shall allow Admin to configure system-level parameters such as borrowing duration limits, per-day fine rates, and membership validity periods	High
FR-AD-07	The system shall maintain a complete audit trail of all administrative actions with timestamp and user identity for accountability purposes	High
FR-AD-08	The system shall allow Admin to broadcast library announcements to all active members via Email	Low

3.2 Library Management Module

3.2.1 Description

This module shall manage the complete book catalogue including addition of new books, classification, and inventory management across all NMC library branches.

3.2.2 Functional Requirements

Req. ID	Requirement Description	Priority
FR-LM-01	The system shall allow librarians to add new books using ISBN-based auto-population of book details including title, author, publisher, and edition from a standard book database	High
FR-LM-02	The system shall support manual entry of book details in cases where ISBN lookup is unavailable or unsuccessful	High
FR-LM-03	The system shall allow classification of books by category, subject, language, and publication year	High
FR-LM-04	The system shall maintain real-time inventory counts for each book title showing total copies, available copies, and currently issued copies	High
FR-LM-05	The system shall support management of books across multiple branch libraries with branch-specific stock levels	High
FR-LM-06	The system shall allow soft deletion (deactivation) of books which are lost, damaged, or permanently withdrawn from circulation	Medium
FR-LM-07	The system shall generate printable barcode or QR code labels for each book to enable scanning during issue and return operations	Medium

3.3 Member Management Module

3.3.1 Description

This module shall handle registration of citizens as library members, generation of unique membership IDs, management of membership validity, and enforcement of borrowing privileges.

3.3.2 Functional Requirements

Req. ID	Requirement Description	Priority
FR-MM-01	The system shall allow librarians to register new members by capturing personal details including full name, residential address, Aadhaar number, mobile number, Email ID, and passport-size photograph	High
FR-MM-02	The system shall auto-generate a unique Membership ID for every registered member following a standardised format (e.g., NMC-LIB-2025-00001)	High
FR-MM-03	The system shall support configurable membership validity periods such as 1 year or 2 years as per NMC policy	High

Req. ID	Requirement Description	Priority
FR-MM-04	The system shall dispatch a Welcome SMS and Email to the member upon successful registration confirming membership details	Medium
FR-MM-05	The system shall enforce configurable borrowing limits per membership tier (e.g., standard members may borrow up to 3 books, premium members up to 5 books simultaneously)	High
FR-MM-06	The system shall allow renewal of memberships prior to or after expiry with applicable renewal fee recording	High
FR-MM-07	The system shall automatically flag and restrict borrowing privileges for members who have overdue books pending return or unpaid outstanding fines	High
FR-MM-08	The system shall provide complete member-wise borrowing history accessible to librarians and to the respective member via their self-service account	Medium

3.4 Book Issue and Return Module

3.4.1 Description

This module shall manage the complete lifecycle of physical book borrowing including issuance, return, reservation queue management, and automated fine calculation for delayed returns.

3.4.2 Functional Requirements

Req. ID	Requirement Description	Priority
FR-IR-01	The system shall allow librarians to issue books to valid members by scanning or manually entering the book barcode and member ID	High
FR-IR-02	The system shall automatically calculate and display the due return date at the time of issuance based on the configured borrowing duration	High
FR-IR-03	The system shall record book returns, update the inventory in real time, and automatically compute overdue fine (if applicable) as per the configured fine rate per day	High
FR-IR-04	The system shall allow registered citizens to place online reservations on book titles which are currently issued to other members	Medium
FR-IR-05	The system shall notify the reserving member via SMS and Email when the reserved book becomes available for collection	Medium

Req. ID	Requirement Description	Priority
FR-IR-06	The system shall maintain a comprehensive transaction log for every book issuance and return activity for audit and reporting purposes	High
FR-IR-07	The system shall generate a printable issue slip or receipt for the member upon successful book issuance	Low
FR-IR-08	The system shall allow librarians to record lost book incidents and levy applicable replacement charges on the concerned member's account	Medium

3.5 Digital Content Library Module

3.5.1 Description

This module shall provide a centralised digital repository for learning resources including eBooks, research papers, study materials, and video content, accessible online by registered members without the need for physical borrowing.

3.5.2 Functional Requirements

Req. ID	Requirement Description	Priority
FR-DC-01	The system shall allow Admins and Library Administrators to upload digital content in formats including PDF, EPUB, and MP4 video files	High
FR-DC-02	The system shall categorise digital content by resource type: eBook, Research Paper, Study Material, or Video Lecture	High
FR-DC-03	The system shall allow registered and active members to browse and access digital content online without any physical borrowing process	High
FR-DC-04	The system shall enforce access control ensuring only valid, active members can access premium digital resources	High
FR-DC-05	The system shall maintain a digital archive with version management capability to accommodate updated editions of existing documents	Medium
FR-DC-06	The system shall track and record digital content access statistics per resource for analytics and utilisation reporting	Medium

3.6 Online Search Portal

3.6.1 Description

The Online Search Portal shall be a publicly accessible citizen-facing interface allowing registered and non-registered users to search the library catalogue. Registered members shall additionally be able to place reservations and access their account details.

3.6.2 Functional Requirements

Req. ID	Requirement Description	Priority
FR-SP-01	The system shall provide a public-facing book search interface accessible without any login requirement for general citizens to browse the physical book catalogue	High
FR-SP-02	The system shall support search by book title, author name, ISBN, subject, and category with keyword auto-suggest functionality	High
FR-SP-03	The system shall display real-time availability status for each book (Available / Issued / Reserved) prominently in search results	High
FR-SP-04	The system shall allow logged-in members to place online reservation requests for books that are currently unavailable	Medium
FR-SP-05	The system shall allow logged-in members to view their currently issued books, applicable due dates, and active reservation status via a self-service account dashboard	High
FR-SP-06	The system shall allow logged-in members to browse and access the Digital Content Library from the portal	High

3.7 Notification Module

3.7.1 Functional Requirements

Req. ID	Trigger Event	Channel	Recipients
FR-NT-01	Successful new member registration	SMS + Email	Registered Member
FR-NT-02	Book issued successfully	SMS	Member
FR-NT-03	Book due date approaching (3 days prior)	SMS + Email	Member
FR-NT-04	Book overdue (daily reminder until returned)	SMS	Member
FR-NT-05	Reserved book now available for collection	SMS + Email	Member
FR-NT-06	Membership expiry approaching (30 days prior)	SMS + Email	Member
FR-NT-07	Library announcements and upcoming events	Email	All Active Members

3.8 Reports and Analytics Module

3.8.1 Functional Requirements

Req. ID	Report Name	Description	Access Level
FR-RP-01	Daily Issue / Return Summary	Date-wise count of books issued and returned per branch	Librarian, Admin
FR-RP-02	Overdue Books Report	List of all books not returned beyond due date with member details and accumulated fine amount	Librarian, Admin
FR-RP-03	Most Borrowed Books Report	Top-N most frequently borrowed book titles over a selected time period	Admin
FR-RP-04	Member Activity Report	Complete borrowing history and current activity status of individual members	Admin, Librarian
FR-RP-05	Library Inventory Report	Current stock levels including available copies, issued copies, and books marked as damaged or lost	Admin, Librarian
FR-RP-06	Fine Collection Report	Total fines levied and collected, filterable by branch and date range	Admin
FR-RP-07	Digital Content Access Report	Utilisation statistics showing download counts and online access frequency per digital resource	Admin
FR-RP-08	Branch Utilisation Comparison	Comparative analytics of membership growth and borrowing activity across all NMC library branches	Super Admin

4. Non-Functional Requirements

4.1 Performance Requirements

Parameter	Requirement
Page Load Time	All primary pages shall load within 3 seconds on a standard broadband connection of minimum 10 Mbps
Concurrent Users	The system shall support a minimum of 200 concurrent users without degradation in response time or functionality
Search Response Time	Book catalogue search results shall be returned within 2 seconds for any valid query
Report Generation	Standard MIS reports shall be generated and rendered within 10 seconds
File Upload	Digital content file uploads of up to 50 MB shall be supported without request timeout
System Uptime	The system shall maintain a minimum uptime of 99.5% during operational hours (6:00 AM to 11:00 PM IST)

4.2 Security Requirements

- All data transmission between the client browser and server shall be encrypted using SSL/TLS (HTTPS protocol mandatory)
- User authentication shall be implemented using JWT (JSON Web Token) based stateless session management
- User passwords shall be stored as salted hashes using the bcrypt algorithm; plain-text storage of passwords is strictly prohibited
- Role Based Access Control (RBAC) shall strictly enforce module-level and action-level permissions for every user role
- All failed login attempts shall be logged; user accounts shall be temporarily locked after 5 consecutive failed login attempts to prevent brute-force attacks
- OTP-based two-factor authentication shall be made available for Admin and Librarian roles
- Automated database backups shall be configured to run at least once every 24 hours with offsite or cloud-based backup storage
- Sensitive citizen data including Aadhaar number and mobile number shall be masked in UI display and restricted from bulk export by unauthorised roles

4.3 Usability Requirements

- The system shall provide a fully responsive user interface compatible with screen sizes ranging from 360px width (mobile) to 1920px (desktop)
- The interface shall be made available in both English and Marathi languages to serve all citizens of Nagpur
- All data entry forms shall include inline validation with clear, user-friendly error messages written in simple language

- The citizen-facing online search portal shall be designed for ease of use by non-technical members of the public without requiring any prior training
- The admin and librarian interfaces shall follow consistent navigation patterns to minimise the number of steps required to complete routine day-to-day operations

4.4 Maintainability and Scalability

- The system shall follow a modular, layered architecture to enable independent updates or enhancements to individual modules without impacting others
- All backend APIs shall be documented using the Swagger / OpenAPI standard to facilitate future integrations or development of a mobile application
- The database schema shall be designed to easily accommodate additional library branches, new book categories, and new types of digital content without requiring structural modifications
- Cluematrix shall provide 1 year of technical support post go-live, inclusive of bug fixes, minor enhancements, and security patch releases

PART II

High Level Design (HLD) Document

5. System Architecture Overview

5.1 Architectural Pattern

The DLMS shall follow a Three-Tier Client-Server Architecture comprising a Presentation Tier, Application (Business Logic) Tier, and Data Tier. The application shall be built on a RESTful API design ensuring clean separation of concerns, independent scalability of tiers, and ease of future integration with third-party systems or mobile applications.

5.2 Architecture Layers

Layer	Technology / Component	Responsibility
Presentation Tier (Frontend)	ReactJS, HTML5, Bootstrap 5, Axios HTTP Client	Renders the user interface, handles user interactions, and communicates with the backend exclusively via REST APIs
Application Tier (Backend)	Java 17, Spring Boot 3.x, Spring Security, Spring Data JPA, Spring Async	Houses all business logic, manages authentication and authorisation, exposes REST API endpoints, handles notifications
Data Tier (Database)	MySQL 8.x or PostgreSQL 15	Persistent and reliable storage of all application data including members, books, transactions, fines, and digital content metadata
File Storage	Server File System or AWS S3-compatible Object Storage	Secure storage of digital content files (PDFs, eBooks, MP4 video resources, member photographs)
Notification Gateway	SMS: MSG91 / Textlocal; Email: SMTP / SendGrid	Reliable delivery of automated SMS and Email notifications to library members and staff
Security Layer	SSL/TLS, JWT, Spring Security, bcrypt Password Hashing	End-to-end encryption, user authentication, role-based authorisation, and secure session lifecycle management

5.3 Deployment Architecture

The following deployment configuration is recommended for the production environment:

- Web Server / Reverse Proxy: NGINX – responsible for HTTPS termination, request routing, load balancing, and serving of static frontend assets
- Application Server: Apache Tomcat (embedded within Spring Boot executable JAR) – hosts and serves the backend REST API application
- Database Server: Dedicated MySQL or PostgreSQL instance with automated daily backup jobs configured
- File Storage: Dedicated storage volume or cloud object storage bucket for digital content and member photographs
- Environment Segregation: Separate Development, User Acceptance Testing (UAT), and Production environments shall be maintained throughout the project lifecycle

6. High Level Module Design

6.1 Module Interaction Overview

The following describes the key interactions between system modules in typical usage scenarios:

- Citizen accesses Online Search Portal, searches the Book Catalogue, places a Book Reservation, and receives an SMS/Email notification upon book availability
- Librarian logs into the Librarian Interface, performs Book Issue or Return operations, the system updates Inventory in real time, and the Notification Service dispatches an SMS confirmation to the member
- Library Administrator manages the Book Catalogue, Member records, fine collections, and views Branch-level MIS Reports via the Admin Dashboard
- Super Administrator governs all library branches, configures system parameters (fine rates, borrowing limits), and views system-wide consolidated analytics
- Background Scheduled Jobs run daily to trigger Due Date Reminders (3 days prior), Overdue Book Alerts, and Membership Expiry Notifications automatically without manual intervention

6.2 Component-wise Design

6.2.1 Authentication and Authorisation Component

- Manages login for all user roles: Super Admin, Library Admin, Librarian, and Citizen Member
- Issues signed JWT access tokens upon successful verification of credentials (token expiry: 8 hours for staff, 24 hours for members)
- Validates role-based permissions at the API gateway level before processing any business request
- Provides OTP-based self-service password reset via the registered mobile number
- Maintains a server-side token blacklist to support immediate session invalidation on logout

6.2.2 Book Catalogue Component

- REST APIs: POST /api/v1/books (add new book), GET /api/v1/books (search and list), PUT /api/v1/books/{id} (edit), DELETE /api/v1/books/{id} (deactivate)
- ISBN lookup integration with Open Library API or Google Books API for auto-population of book metadata during entry
- Real-time inventory counter maintained via application-level transactional updates ensuring accuracy of stock levels at all times

6.2.3 Member Management Component

- REST APIs: POST /api/v1/members (register), GET /api/v1/members/{id} (view), PUT /api/v1/members/{id} (update), POST /api/v1/members/{id}/renew (renew membership)
- Membership ID auto-generated following a standardised format with branch prefix: e.g., NMC-LIB-2025-00001

- Member photograph and Aadhaar document uploaded via secure multipart file upload and stored in the designated file storage layer

6.2.4 Issue and Return Transaction Component

- REST APIs: POST /api/v1/transactions/issue, POST /api/v1/transactions/return, GET /api/v1/transactions/member/{memberId}
- Fine Calculation Formula: Fine Amount = (Actual Return Date – Configured Due Date) × Configured Fine Rate Per Day
- Reservation queue management follows FIFO (First In, First Out) policy ensuring fairness in reservation fulfilment
- Inventory count updated atomically within the same database transaction as issue/return to prevent data inconsistency

6.2.5 Digital Content Component

- Multipart file upload API with server-side validation of file type (PDF, EPUB, MP4 permitted) and file size (maximum 50 MB per upload)
- Video content served via HTTP byte-range streaming to enable in-browser playback without requiring full download
- Access control middleware validates active membership status before serving any premium digital resource

6.2.6 Notification Service Component

- Asynchronous notification dispatch implemented using Spring @Async or a lightweight message queue (RabbitMQ) to avoid blocking the main application thread
- Configurable notification message templates stored in the database for easy customisation by Admin without requiring code changes
- Complete notification delivery log maintained for each dispatch attempt, including delivery status and retry history, for audit purposes

6.2.7 Reporting and Analytics Component

- All reports generated via optimised parameterised SQL queries with support for date-range filters and branch-level filters
- Report export functionality provided in both PDF (for printing) and Excel (.xlsx) formats for all standard MIS reports
- Dashboard summary charts and trend graphs rendered on the frontend using Chart.js consuming aggregated data from dedicated reporting APIs

7. High Level Database Design

7.1 Overview

The DLMS database shall follow a normalised relational schema at minimum Third Normal Form (3NF) to ensure data integrity, eliminate data redundancy, and support efficient query execution. All primary keys shall use auto-incremented integer or UUID identifiers. Foreign key constraints shall be enforced at the database level to maintain referential integrity across all related entities.

7.2 Key Entities and Table Design

Table Name	Key Columns	Description
tbl_branches	branch_id (PK), branch_name, address, city, pincode, contact_number, status, created_at	Master record of all NMC library branches
tbl_users	user_id (PK), name, email, mobile, password_hash, role (ENUM: SUPER_ADMIN/LIB_ADMIN/LIBRARIAN), branch_id (FK), status, last_login, created_at	System user accounts for administrative staff
tbl_books	book_id (PK), isbn, title, author, publisher, edition, category, subject, language, total_copies, available_copies, branch_id (FK), status, created_at	Master book catalogue with branch-wise inventory counts
tbl_members	member_id (PK), membership_no (UNIQUE), name, mobile, email, aadhaar_no (encrypted), photo_path, membership_type, valid_from, valid_to, branch_id (FK), status, created_at	Registered citizen library members
tbl_transactions	txn_id (PK), member_id (FK), book_id (FK), issued_by (FK to tbl_users), issued_date, due_date, return_date, returned_by (FK), fine_amount, fine_paid (BOOL), status (ENUM: ISSUED/RETURNED/LOST)	Book issuance and return transaction ledger
tbl_reservations	reservation_id (PK), member_id (FK), book_id (FK), reserved_on, queue_position, status (ENUM: PENDING/FULFILLED/CANCELLED/EXPIRED), notified (BOOL)	Book reservation queue
tbl_digital_content	content_id (PK), title, author, category, content_type (ENUM: EBOOK/VIDEO/RESEARCH_PAPER/STUDY_MATERIAL), file_path, file_size_mb, uploaded_by (FK), uploaded_at, access_count, status	Digital content repository metadata

Table Name	Key Columns	Description
tbl_notifications	notif_id (PK), member_id (FK), trigger_event, channel (ENUM: SMS/EMAIL), message_body, status (ENUM: SENT/FAILED/PENDING), sent_at, retry_count	Notification dispatch log
tbl_fines	fine_id (PK), txn_id (FK), member_id (FK), fine_amount, amount_paid, payment_mode, payment_date, collected_by (FK), status (ENUM: PENDING/PARTIAL/PAID)	Fine levy and collection tracking
tbl_system_config	config_key (PK), config_value, data_type, description, updated_by (FK), updated_at	System-level configurable parameters such as fine rate, borrowing limit, and membership validity

7.3 Key Relationships

- tbl_branches (1) to (Many) tbl_books – Each library branch maintains its own book inventory
- tbl_branches (1) to (Many) tbl_members – Each member is associated with a primary library branch
- tbl_members (1) to (Many) tbl_transactions – A member may have multiple book borrowing records over their membership period
- tbl_books (1) to (Many) tbl_transactions – Each book title may have multiple borrowing transaction records across its lifetime
- tbl_members (1) to (Many) tbl_reservations – A member may hold reservations on multiple book titles simultaneously
- tbl_transactions (1) to (0 or 1) tbl_fines – An overdue transaction generates at most one fine record

8. High Level API Design

8.1 API Design Principles

- All APIs shall conform to RESTful conventions using standard HTTP methods: GET for retrieval, POST for creation, PUT for update, DELETE for deactivation
- All request and response payloads shall use JSON (application/json) as the data interchange format
- All protected API endpoints shall require a valid JWT Bearer token in the Authorization header of every request
- API versioning shall be enforced via URL prefix to facilitate backward-compatible upgrades (e.g., /api/v1/books)
- A standardised error response envelope shall be returned for all failures: { status, errorCode, message, timestamp }

8.2 Key API Endpoints Summary

Module	Method	Endpoint	Description
Auth	POST	/api/v1/auth/login	User login – validates credentials and returns signed JWT token
Auth	POST	/api/v1/auth/logout	Invalidates the current user session token
Auth	POST	/api/v1/auth/forgot-password	Initiates OTP-based self-service password reset
Books	GET	/api/v1/books	Search and list books with filters for title, author, ISBN, category
Books	POST	/api/v1/books	Add a new book to the catalogue (Librarian and above)
Books	PUT	/api/v1/books/{id}	Update book details or adjust inventory count
Books	GET	/api/v1/books/{id}/availability	Check real-time copy availability for a specific book
Members	POST	/api/v1/members	Register a new library member
Members	GET	/api/v1/members/{id}	Retrieve member profile and borrowing history
Members	PUT	/api/v1/members/{id}/renew	Process membership renewal with updated validity dates
Transactions	POST	/api/v1/transactions/issue	Issue a book to a registered member
Transactions	POST	/api/v1/transactions/return	Record book return and compute overdue fine if applicable

Module	Method	Endpoint	Description
Reservations	POST	/api/v1/reservations	Place a reservation request for an unavailable book
Digital Content	GET	/api/v1/digital-content	List all available digital resources with filters
Digital Content	POST	/api/v1/digital-content/upload	Upload a new digital resource file (Admin only)
Reports	GET	/api/v1/reports/overdue	Fetch overdue books report with filter support
Reports	GET	/api/v1/reports/inventory	Fetch current branch-wise inventory status report
Reports	GET	/api/v1/reports/member-activity	Fetch member-wise borrowing activity report
Reports	GET	/api/v1/reports/fines	Fetch fine collection report filterable by branch and date

9. Security Design

9.1 Authentication Flow

- Step 1: User submits credentials (username and password) via HTTPS POST to /api/v1/auth/login
- Step 2: Backend retrieves the stored bcrypt password hash for the provided username and validates the submitted password against it
- Step 3: On successful validation, the server generates and returns a signed JWT access token with embedded role and branch information
- Step 4: The client application stores the JWT in memory (not in browser localStorage or cookies) and attaches it as a Bearer token in the Authorization header of all subsequent API requests
- Step 5: On logout or token expiry, the token is added to a server-side blacklist cache (Redis or in-memory store) to prevent reuse of invalidated tokens

9.2 Role-wise Authorisation Matrix

Module / Action	Super Admin	Library Admin	Librarian	Member
Add / Edit Library Branch	Yes	No	No	No
Manage Staff User Accounts	Yes	No	No	No
Add / Edit / Deactivate Books	Yes	Yes	Yes	No
Register New Members	Yes	Yes	Yes	No
Issue and Return Books	Yes	Yes	Yes	No
View Own Borrowing History	No	No	No	Yes
Place Online Book Reservation	No	No	No	Yes
Access Digital Content	Yes	Yes	Yes	Yes (active members)
Upload Digital Content	Yes	Yes	No	No
View MIS Reports	Yes (all branches)	Yes (own branch)	Yes (own branch)	No
Configure System Parameters	Yes	No	No	No
View Audit Trail Logs	Yes	No	No	No

10. Implementation Roadmap

Phase	Activities	Duration	Key Deliverables
Phase 1 – Requirement Finalisation	Detailed requirement workshops with NMC IT Department, sign-off on finalised SRS document, development environment and repository setup	Week 1	Approved SRS Document, Dev Environment Setup
Phase 2 – UI/UX Design	Wireframe creation for all modules, clickable interactive prototype for citizen portal and admin dashboard, design review and formal approval by NMC	Week 1	Approved UI Wireframes, Interactive Prototype
Phase 3 – Development Sprint 1	Database schema and data model creation, authentication module, admin dashboard, book catalogue module, member management module, core REST APIs	Week 1-2	Backend APIs (Sprint 1), Admin Module UI
Phase 4 – Development Sprint 2	Book issue/return module, reservation system, fine calculation engine, digital content repository, online citizen search portal, notification service integration	Weeks 3-5	Complete Functional Application
Phase 5 – Integration Testing and UAT	End-to-end system integration testing, user acceptance testing with NMC library staff, bug fixing and regression testing, performance and basic security testing	Weeks 6-7	UAT Sign-off, Test Reports, Bug Fix Releases
Phase 6 – Deployment and Training	Production server configuration, SSL certificate installation, data migration (if applicable), library staff and admin training sessions, go-live and hypercare	Week 8	Live Production System, Trained Staff, User Manuals

11. Risks and Mitigation Plan

Risk	Likelihood	Impact	Mitigation Strategy
Delay in server infrastructure procurement by NMC	Medium	High	Cluematrix shall provision a temporary cloud environment for development and UAT at no additional cost to NMC to ensure project timelines are not impacted
Incomplete or inconsistent existing library data for migration	High	Medium	A structured data cleansing exercise shall be conducted during Phase 1. Cluematrix shall provide standardised data migration templates and validation tools
Low adoption by library staff due to resistance to change	Medium	High	Comprehensive hands-on training sessions, Marathi and English user manuals, and a 3-month hypercare support period with dedicated on-call assistance shall be provided post go-live
Third-party SMS gateway service disruption	Low	Medium	The system shall be integrated with a secondary SMS gateway as a failover. Email notifications shall serve as the backup channel during SMS service disruption
Security vulnerabilities discovered in production	Low	High	A mandatory VAPT (Vulnerability Assessment and Penetration Testing) exercise shall be conducted prior to go-live. All critical and high-severity findings shall be remediated before the system is made live
Scope creep leading to timeline overrun	Medium	Medium	Any new requirements raised after SRS sign-off shall be documented as a Change Request (CR) with a separate effort estimate and timeline, ensuring the original scope and go-live date are protected

12. Glossary of Terms

Term	Meaning
DLMS	Digital Library Management System – the integrated software platform described throughout this document
NMC	Nagpur Municipal Corporation – the client organisation and the governing body responsible for all public libraries covered under this project
ISBN	International Standard Book Number – a globally unique 13-digit numeric identifier assigned to each published book edition
JWT	JSON Web Token – a compact and self-contained mechanism for securely transmitting user authentication and authorisation information between the client and server
RBAC	Role Based Access Control – a security access model wherein permissions are assigned to roles rather than individual users, and users are then assigned to appropriate roles
REST API	Representational State Transfer Application Programming Interface – a standardised architectural style for designing networked application interfaces using HTTP
UAT	User Acceptance Testing – a testing phase in which actual end-users (NMC library staff) validate the system against their defined requirements and real-world usage scenarios
VAPT	Vulnerability Assessment and Penetration Testing – a structured security testing methodology to systematically identify, verify, and remediate system vulnerabilities before go-live
MIS	Management Information System – refers to the reporting, analytics, and decision-support capabilities of the DLMS used by library administrators and NMC management
Hypercare	An intensive post-deployment support period immediately following go-live during which the Cluematrix team provides elevated monitoring, rapid response to issues, and close end-user hand-holding
Change Request (CR)	A formally documented request for a modification to the agreed project scope, subject to separate effort estimation, cost assessment, and timeline revision
FIFO	First In, First Out – the queue management policy applied to book reservations ensuring the member who reserved earliest receives the book first upon its return

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